

Corexion Academy API Documentation

Overview

Corexion Academy is an online learning platform API built using:

- ASP.NET Core Web API
- Entity Framework Core
- SQL Server
- JWT Authentication
- ASP.NET Identity
- Generic CRUD Services
- Role-Based Authorization

The system supports:

- Authentication & Authorization
 - Courses Management
 - Course Sections
 - Lessons & Video Streaming
 - Enrollments
 - Payments
 - Reviews
 - Quizzes & Student Results
-

Architecture

The project follows a layered architecture:

Layers

API Layer

Contains:

- Controllers
- Middleware
- Request handling
- Authorization

Application Layer

Contains:

- DTOs
- Interfaces
- Business contracts
- Application services

Infrastructure Layer

Contains:

- Service implementations
- Repository logic
- JWT implementation
- File services
- Database access

Domain Layer

Contains:

- Entities
 - Enums
 - Core business models
-

Authentication System

Authentication Type

JWT Bearer Authentication.

Roles

The platform supports:

- Admin
- Instructor
- Student

Features

- Register
- Login

- Email Confirmation
 - Forgot Password
 - Reset Password
 - Refresh Token
 - Logout
 - Admin Create User
-

Base API URL

/api

Authentication Endpoints

Register

Endpoint

POST /api/Auth/register

Request Body

```
{
  "fullName": "Mahmoud Ashraf",
  "email": "test@test.com",
  "password": "P@ssw0rd"
}
```

Response

```
{
  "success": true,
  "message": "Registration successful"
}
```

Login

Endpoint

```
POST /api/Auth/login
```

Request

```
{
  "email": "test@test.com",
  "password": "P@ssw0rd"
}
```

Response

```
{
  "success": true,
  "token": "JWT_TOKEN",
  "refreshToken": "REFRESH_TOKEN",
  "role": "Student",
  "expiration": "2026-01-01T00:00:00"
}
```

Confirm Email

```
GET /api/Auth/confirm-email
```

Forgot Password

```
POST /api/Auth/forgot-password
```

Reset Password

```
POST /api/Auth/reset-password
```

Refresh Token

POST /api/Auth/refresh

Logout

POST /api/Auth/logout

Courses Module

Features

- Create Course
- Update Course
- Delete Course
- Get All Courses
- Get Course By Id
- Change Course State

Course Status

- Pending
- Approved
- Rejected

Get All Courses

GET /api/Courses/GetAllCourses

Roles

- Instructor
 - Admin
 - Student
-

Create Course

`POST /api/Courses/CreateCourse`

Roles

- Instructor
- Admin

Request Example

```
{
  "title": "ASP.NET Core",
  "description": "Complete ASP.NET Course",
  "price": 100
}
```

Update Course

`PUT /api/Courses/UpdateCourse/{id}`

Delete Course

`DELETE /api/Courses/DeleteCourse/{id}`

Change Course State

`POST /api/Courses/SetState`

Request

```
{
  "id": 1,
```

```
"state": "active"
}
```

Course Sections Module

Features

- Create Section
- Update Section
- Delete Section
- Get All Sections
- Get Section By Id

Create Course Section

```
POST /api/CourseSections/CreateCourseSection
```

Roles

- Instructor
- Admin

Lessons Module

Features

- Create Lesson
- Upload Video
- Stream Video
- Delete Video
- Update Lesson

Create Lesson

```
POST /api/lesson/create
```

Content Type

multipart/form-data

Roles

- Instructor
 - Admin
-

Upload Video

POST /api/lesson/upload-video/{id}

Stream Video

GET /api/lesson/stream/{id}

Roles

- Instructor
 - Admin
 - Student
-

Enrollments Module

Features

- Enroll Student
 - Update Enrollment
 - Delete Enrollment
 - Get Enrollments
-

Create Enrollment

POST /api/Enrollments/CreateEnrollment

Payments Module

Features

- Create Payment
- Get Payments
- Payment Tracking

Create Payment

`POST /api/Payments/CreatePayment`

Reviews Module

Features

- Add Review
- Update Review
- Delete Review
- Get Reviews

Create Review

`POST /api/Reviews/CreateReview`

Quizzes Module

Features

- Create Quiz
 - Update Questions
 - Submit Quiz
 - Calculate Results
-

Create Quiz

```
POST /api/Quizzes/CreateQuiz
```

Roles

- Instructor
 - Admin
-

Submit Quiz

```
POST /api/Quizzes/{quizId}/submit
```

Roles

- Student
-

File Storage System

Features

- Course folders creation
- Section folders creation
- Video upload
- Video replacement
- Video streaming
- Video deletion

Storage Structure

```
PrivateVideos/  
  CourseName/  
    SectionName/  
      LessonVideo.mp4
```

Generic CRUD System

Supported Operations

- Create
- Update
- Delete
- Get By Id
- Get All

Features

- Reflection-based mapping
 - Foreign key validation
 - Unique index validation
 - Dynamic includes
-

Authorization Matrix

Module	Admin	Instructor	Student
Courses	Full	Own Courses	View Approved
Lessons	Full	Own Lessons	View
Quizzes	Full	Create & Edit	Submit
Reviews	Full	View	Create
Payments	Full	View	Pay
Enrollments	Full	View	Enroll

Database Technologies

- Entity Framework Core
 - SQL Server
 - ASP.NET Identity
-

Security

Implemented

- JWT Authentication
- Role Authorization
- Refresh Tokens
- Password Hashing
- Identity Framework

Planned Improvements

- Ownership Authorization
 - Secure Video Access
 - Refresh Token Rotation
 - Global Exception Middleware
 - AutoMapper Integration
 - Better Validation Layer
-

Response Wrapper

The API uses a generic response wrapper.

Example

```
{
  "success": true,
  "message": "Operation completed successfully",
  "data": {}
}
```

Error Response

```
{
  "success": false,
  "message": "Validation failed"
}
```

Future Enhancements

Planned Features

- Live Streaming
 - Certificates
 - Notifications
 - Course Progress Tracking
 - AI Recommendations
 - Payment Gateway Integration
 - Swagger Documentation Enhancements
 - Dashboard Analytics
-

Project Status

Current Status:

- Core Backend Completed
 - Authentication Completed
 - Video System Completed
 - Quiz System Completed
 - CRUD Modules Completed
 - Security Hardening In Progress
 - Production Optimization Pending
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Developed Using

- ASP.NET Core 8
 - Entity Framework Core
 - SQL Server
 - JWT
 - Identity
 - LINQ
 - C#
-

Notes

This documentation represents the current backend implementation state. Some security optimizations and production improvements are planned for future updates.